

Ilja Safronovs

R&D Engineer — 3D reconstruction · AI automation · Giraffe360 · Riga, Latvia

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I work on 3D reconstruction and AI automation. In 3D: mainly Gaussian splatting, and a bit of meshes. At Giraffe360 I build our splat pipeline from sparse panoramas and implement published methods in PyTorch and CUDA. In production: the windows and mirrors fix, my densification changes, the exterior orbital videos and the looped drone. On the automation side I build tools that help the team build faster: an experiment runner driven by AI agents, an MCP server for the team task board and a coding-agent harness. I also work on depth models, 3D camera paths and LLM APIs.

EXPERIENCE

Giraffe360 — R&D Engineer — Riga · 2025 – present

Fixed blurry windows, outdoor views and mirrors in sparse-panorama 3DGS; the production default since 1 June 2026. instagram.com/p/DZsNRB8o772

I ran 140+ controlled training runs, then 600+ experiments over one month to cut haze, floaters and flicker.

Built our 3DGS pipeline from sparse panoramas, with my own densification and CUDA kernels.

The pipeline turns sparse panorama views into 3D Gaussian splats. Jan 2026: my influence-accumulation densification, building on Taming 3DGS (Mallick et al., 2024). June 2026: Analytic-Splatting (Liang et al., 2024) in gsplat, with my CUDA forward and backward passes.

Shipped two 3D video features: exterior orbital videos and the looped drone (Oct 2026).

instagram.com/giraffe360/reel/Dd9ODBVIHhE

I led the exterior work. I built the toolkit that orients, selects and crops the outdoor splat, and a per-frame video pipeline; it makes the orbital view in production. The looped drone was my hackathon idea (Feb 2026, special mention); I built its camera-path algorithm.

Mentored a junior engineer on a new depth method; the whole team uses it.

Mostly his work: minima matching plus OMNI-DC to densify LiDAR depth.

Built tools that help the team build faster: quality checks, an experiment runner, one monorepo.

July 2026: ground-truth datasets and validation tools for splat quality. March 2026: an experiment runner, so every teammate can run and track training on a local GPU through an AI agent.

Built an embeddable Rust/wgpu splat viewer and a comparison tool.

A Rust/wgpu viewer (May 2026) that any app can embed and control, and a multi-view comparison tool (Jan 2026).

Prototypes: mirror reflections, meshes from splats, lighter splats, a fly-in video intro.

Working prototypes: correct reflections in mirrors (June 2026), mesh generation from splats, a stage that cuts the splat count for viewers (May 2026), and a fly-in video intro.

Agent tools: an MCP server for the team task board, and a coding-agent harness.

June 2026: an MCP server (Node.js), so AI agents can read and write the team task board; colleagues use it.

Jul–Oct 2025: a small LLM service that wrote tour narration.

It used the OpenAI API and AWS S3.

Giraffe360 makes cameras and software for real-estate virtual tours.

EDUCATION

Riga Technical University — Professional Bachelor, Financial Engineering — 2024 – 2028 (expected)

Probability, statistics, stochastic analysis, Monte Carlo methods, algorithms, databases.

Qualification: senior data analysis specialist. Other coursework: mathematical analysis and algebra, econometrics, time-series analysis, portfolio analysis, actuarial mathematics, computer networks, programming (Java, C++).

AI Talent Program (Ponder) — Nov 2024 – Jan 2025

Mākslīgā intelekta talantu programma

C++, Python, maths and AI with industry mentors; it led to the Giraffe360 role.
A selective program. Theory of AI: its history, neural network architectures (CNNs, transformers, GANs, VAEs), computer vision and reinforcement learning.

Riga State Classical Gymnasium — Programming and Mathematics track
Rīgas Valsts klasiskā ģimnāzija

State mathematics olympiads and the Small Mathematics University.
Specialised track in programming and mathematics. Latvian state mathematics olympiads; completed the Small Mathematics University (Mazā matemātikas universitāte) at the University of Latvia.

PROJECTS

Portfolio site: a walkable Gaussian splat of a real flat that I trained (gsplat, 730k splats).

sort_visualization — 16 sorting algorithms in C++/SFML; runs in the browser (WebAssembly).

Offline Graphing Calculator — 4th place of 27 in a school competition.

monte-carlo-finance — option pricing, volatility and risk models in Python.

load-balancing-experiments — seven experiments on DNS, queueing and load balancing.

flow — offline workout tracker in Dart/Flutter, with CI and golden-image tests.

crypto-quant-lab — 59 backtested trading-rule versions on 6 coins.

projects_with_NEAT — neuroevolution agents that learn Flappy Bird and Pong.

RTU_program_parser — finds extra RTU lectures that fit your schedule.

Snake + maze generator — maze generation, terrain and path-finding in Pygame.

bezier-curve-js — interactive Bezier-curve drawing tool.

SKILLS

3D & vision	3D Gaussian splatting (gsplat, CUDA kernels), 3D reconstruction from sparse views, computer vision, depth densification, mesh generation, 3D camera paths, splat viewers (Rust/wgpu, three.js/Spark), PyTorch
Research	Reading and implementing papers, controlled experiments, ground-truth evaluation
Languages	Python, C++, Dart/Flutter, JavaScript, Java, SQL
Tools	Linux, Docker, Git/GitLab CI, AWS S3, Pygame, Selenium, pandas
Other	LLM APIs, agent tools (MCP), three.js/WebGL
Spoken	Latvian, English, Russian